

NATIONAL BOARD OF HIGHER EDUCATION

PHYSICS QUALIFYING EXAMINATION

Time Allowed: 3 Hours

Maximum Marks: 100

General Instructions:

1. All questions are compulsory.
2. Read each question carefully before answering.

SECTION A: OBJECTIVE TYPE QUESTIONS

1. What is the SI unit of magnetic flux density?

- A) Weber
- B) Tesla
- C) Henry
- D) Gauss

2. Which electromagnetic wave has the highest frequency?

- (A) Infrared rays
- (B) Ultraviolet rays
- (C) Gamma rays
- (D) Microwaves

Q3. The electric potential inside a hollow charged conducting sphere is:

- 1) Zero
- 2) Constant and equal to potential at surface
- 3) Directly proportional to radius
- 4) Inversely proportional to distance from center

Question 4. Identify whether the following statement is true or false:

An ideal voltmeter has infinite internal electrical resistance.

- A) True
- B) False

Question 5. Explain the physical principles of electromagnetic induction underlying Faraday's second law, particularly regarding the time rate of magnetic flux change through an arbitrary closed surface,

and determine the magnitude of induced electromotive force in a rotating conductive loop.

Also state how Lenz's law ensures the conservation of mechanical energy in electromagnetic systems.

- A) Directly proportional to the rate of change of magnetic flux
- B) Inversely proportional to magnetic flux
- C) Independent of magnetic flux variation
- D) Equal to magnetic vector potential

6. What is the de Broglie wavelength of an electron accelerated through 100 Volts?

- A) 0.123 nm
- B) 1.23 nm
- C) 12.3 nm
- D) 0.012 nm

7. The phenomenon of light bending around sharp obstacles is called:

- (A) Polarization
- (B) Refraction
- (C) Diffraction
- (D) Dispersion

Q8. Fill in the blank:

The work done in moving a test charge over an equipotential surface is always equal to _____.